Mayuri Paliwal

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LinkedIn · Github · LeetCode · Portfolio

Education

Indira Gandhi Delhi Technical University for Women — Delhi, India
B.Tech Electronics and Communication Engineering — Nov 2022 – May 2026
CGPA: 7.8/10

Skills

Languages:	**Python, C++, JavaScript**, Dart
Frontend:	**HTML, CSS, React.js**, Flutter
Backend & Databases:	**FastAPI, REST APIs, SQL, PostgreSQL, Redis**, SQLite, Firebase
CS Fundamentals:	Data Structures and Algorithms, OOP, DBMS, OS
Tools:	**Docker, Git, GitHub Actions, CI/CD**, Pytest, Locust

Work Experience

Alpha AI — Remote
Flutter Developer Intern — Jun 2025 – Aug 2025

- Developed a native Windows audio capture module using **C++ and WASAPI**, handling microphone and system-audio loopback capture, PCM format conversion, buffering, and multi-threaded audio processing.
- Built end-to-end applications for **2 platforms (Android & Desktop)**, implementing **4+ features** powered by open-source speech-to-text and LLM frameworks.
- Engineered and optimized **backend and media processing pipelines** using **FFmpeg, multi-threading, and adaptive model selection**, achieving up to **2× faster inference**.
- Integrated **LLM and speech-to-text APIs** into application workflows and debugged data flows across model inference, backend processing, and client applications.
- Integrated Google Calendar with caching and lifecycle-aware synchronization, reducing redundant API calls by **67%**.

Projects

URL Shortener *React.js, FastAPI, PostgreSQL, Redis, ARQ* — GitHub · Live
Developed and deployed a full-stack URL shortening service with **REST APIs, authentication, Redis caching, asynchronous analytics, and rate limiting**.

- Designed and implemented REST APIs using **Python and FastAPI** for URL shortening, redirection, and analytics, with PostgreSQL persistence and **Base62-encoded database IDs** for short URL generation.
- Implemented **Redis caching** using a cache-aside strategy and **ARQ background jobs** for asynchronous analytics writes, reducing repeated PostgreSQL lookups and allowing redirects to return without waiting for analytics updates.
- Diagnosed redirect latency under concurrent load and introduced **PostgreSQL connection pooling**, reducing **P95 latency from 8.6s to 430ms at 100 concurrent users** in an indicative Locust benchmark.

VeriFace — Aadhaar + Face + Age Verification *React.js, MERN (Team Project)* — GitHub
Built during Zynga Hackathon 2025 for real-time identity verification.

- Created **React.js components** for webcam capture, Aadhaar upload, and live verification progress using **Server-Sent Events (SSE)**, enabling reliable identity verification in **45–60s across 30+ test cases**.
- Collaborated with a **5-member team** integrating frontend workflows with the AI backend using **PyTorch, FaceNet, and multilingual OCR**, contributing to **87%+ match scores** across diverse Aadhaar formats.

Achievements

- **LeetCode:** Solved **600+** DSA problems in C++, achieving a peak contest rating of **1736**.
- **Aven Coding Hackathon 2.0:** Secured **18th rank** in the Coding Contest and received a **Certificate of Excellence**.
- **Microsoft Learn Student Ambassador (MLSA):** Completed structured Microsoft Learn modules on cloud and AI technologies and participated in peer learning initiatives through workshops.